

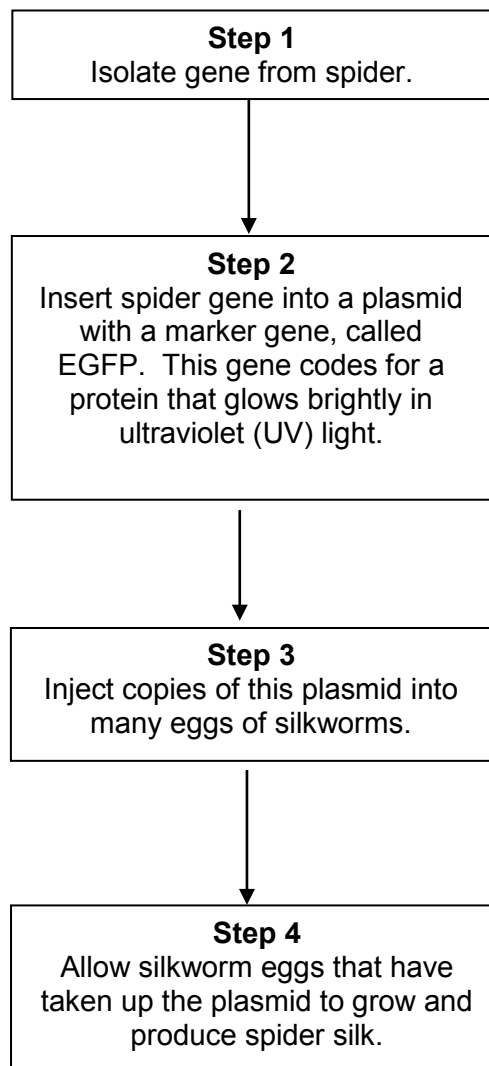
5 Silkworms secrete silk fibres, which are harvested and used to manufacture silk fabric.

Scientists have produced genetically modified (GM) silkworms that contain a gene from a spider.

The GM silkworms secrete fibres made of spider web protein (spider silk), which is stronger than normal silk fibre protein.

The method the scientists used is shown in **Figure 5**.

Figure 5



0 5 . **1** Suggest why the plasmids were injected into the eggs of silkworms, rather than into the silkworms.

[2 marks]

0 5 . **2** Suggest why the scientists used a marker gene and why they used the EGFP gene.

[2 marks]

The scientists ensured the spider gene was expressed only in cells within the silk glands.

0 5 . **3** What would the scientists have inserted into the plasmid along with the spider gene to ensure that the spider gene was only expressed in the silk glands of the silkworms?

[1 mark]

0 5 . **4** Suggest **two** reasons why it was important that the spider gene was expressed only in the silk glands of the silkworms.

[2 marks]

1

2
